

SIMATS ENGINEERING
ASSESSMENT TOOL 2: LABORATORY PERFORMANCE

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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ASSESSMENT TASKS:

4. Develop an AWT application containing two text fields and buttons for: ADD
CLEAR- When the ADD button is clicked, calculate and display the sum of two numbers.
When CLEAR is clicked, clear the input fields." Test Cases:
tc1_input- Number 1=10, Number 2=20, Click ADD
tc1_output- Sum = 30 tc2_input- Number 1=100,
Number 2=250, Click ADD tc2_output- Sum = 350
tc3_input- Number 1=-10, Number 2=20, Click ADD
tc3_output- Sum = 10

CODE:

A. JAVA CODE import

```
java.awt.*;      import
```

```
java.awt.event.*;
```

```
public class AdditionAWT extends Frame implements ActionListener {
```

```
    Label number1Label, number2Label, resultLabel;
```

```
    TextField number1Field, number2Field, resultField;
```

```
    Button addButton, clearButton;
```

```
public AdditionAWT() {

    setTitle("Addition Calculator");
    setSize(450, 300);    setLayout(new
    GridLayout(4, 2, 10, 10));

    number1Label = new Label("Number 1:");
    number1Field = new TextField();

    number2Label = new Label("Number 2:");
    number2Field = new TextField();

    resultLabel = new Label("Result:");
    resultField = new TextField();
    resultField.setEditable(false);

    addButton = new Button("ADD");
    clearButton = new Button("CLEAR");

    addButton.addActionListener(this);
    clearButton.addActionListener(this);

    add(number1Label);
    add(number1Field);    add(number2Label);
    add(number2Field);
```

```

        add(resultLabel);
add(resultField);

        add(addButton);
add(clearButton);

        addWindowListener(new WindowAdapter() {
public void windowClosing(WindowEvent e) {
            System.exit(0);
        }
});

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {

        if (e.getSource() == addButton) {
            try
        {
                double number1 = Double.parseDouble(number1Field.getText());
double number2 = Double.parseDouble(number2Field.getText());           double
                sum = number1 + number2;

```

```
        if (sum == (int) sum) {
resultField.setText("Sum = " + (int) sum);
        } else {
            resultField.setText("Sum = " + sum);
        }

    } catch (NumberFormatException ex) {
        resultField.setText("Invalid Input");
    }
}

if (e.getSource() == clearButton) {
number1Field.setText("");
number2Field.setText("");    resultField.setText("");
    }
}
```

```
public static void main(String[] args) {
new AdditionAWT();
}
}
```

OUTPUT:

 Addition Calculator

Number 1:

50

Number 2:

289

Result

Sum = 339

ADD

CLEAR